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Design and implementation of a user centered design framework for software development

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Abstract: **The concept and methodology known as User-centred Design (UCD) puts the user at the center and concentrates on cognitive aspects as they relate to how people interact with objects. In order to overcome biassed decision-making, automated AI-based decision support systems need to be able to be used, trusted, and accepted. Design criteria for AI-based decision support systems in traditional craftsmanship are therefore required. In this contribution, we examined the relationships between user acceptance and a user-centred design by evaluating three distinct applications in a mixed-method user research (N D = 20) that included both qualitative (think aloud) and quantitative (survey) components. Starting with research-related planning agreements, the context is established by interviewing the parties who agreed in the first stage, the needs are ascertained through user assessments, the System Usability Scale (SUS) method is used to determine the results, a prototype is designed based on the findings of the third step, and UI evaluation is conducted using the same method when evaluating the UI AI running in the third stage to demonstrate whether or not there has been a significant improvement. It took into account technological adoption, usability, and planning objectivity and efficiency. The findings imply that while AI-based decision support systems improve speed and objectivity, usability, acceptance, and confidence depend on user-centred design. Given that automated options were not as often questioned, it makes sense to give the decision maker the last say. It produces practical recommendations for the development of AI-based manufacturing support systems.**

Keywords: User centered design, System usable scale, software development, Usability, Artificial intelligence.

I. Introduction

Although there are numerous methods to describe User-centred Design (UCD), all of them share the emphasis on the user and the incorporation of the user's viewpoint throughout the whole design process. "A philosophy based on the needs and interests of the user, with an emphasis on making products usable and understandable," is how Donald Norman characterises UCD [1]. According to this notion, UCD does not necessarily include actual user engagement. Nonetheless, one popular method of making sure that users' wants and interests are being satisfied is to include them in the User-centred Design process.

At the moment, the development of business and product software is concentrated on how to shorten product delivery times and adapt to changing market demands [2, 3]. One of the most widely used approaches in software development is agile, which aims to address current technology, continuous development, and highly variable user needs [4, 5]. Additionally, 200 organisations had a boost in revenue as a result of applying Agile [6].

UI modifications can be done in a variety of ways. Among these is the User-centred Design (UCD) approach, a development paradigm that emphasises the user's contribution in identifying their needs [7]. This is consistent with the design, which centres the UI prototype creation process around the user. The four guiding concepts of UCD in application development are interactive design, user testing, integrated design, and user focus. Because the UCD approach is highly focused on meeting user requirements for an application and has been shown to yield accurate usability values, the author employs it to address issues from his research.

An evidence-based methodology known as "user-centred design" involves interacting with end users and giving their requirements first priority when creating a service or product [8]. The World Health Organisation (WHO) supports this strategy and suggests incorporating it into the mHealth interventions' lifecycle to guarantee successful results [9], taking usability and functionality into consideration [10]. Therefore, it is necessary to take into account both the technical and social aspects of the app, and how it fits into end users' life. Few frameworks currently exist that enable this comprehensive approach to the building of mHealth apps in underdeveloped and developing countries. Although there are frameworks for designing mHealth apps [11-13], very few of them are made for a developing environment.

The impact of user-centred AI-based decision support systems on user acceptability and usage intention is examined in this work. Two AI-based process recommender systems are created for this goal, one with a user-centred design and the other with a strictly functional one, in order to plan textile reinforced composite processes.

II. Literature Review

A. User Centered Design

The process of designing a user interface that emphasises usability objectives, user attributes, environment, tasks, and workflow is known as user-centred design. UCD analyses, designs, and evaluates popular hardware, software, and online interfaces using a set of well defined processes and approaches. From the very beginning of a project to its completion, design and assessment phases are integrated into the iterative UCD process [14]. "To give early focus on requirements and tasks for Empirical Measurement and testing of product usage and also focusses on iterative Design," is how some studies defines the UCD principles. UCD's objective is to create products with a high level of usefulness. The "extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use" is how some academics define usability [15]. According to studies, usability objectives are: Attitude (likability), Learnability, Effectiveness (ease of use), and Usefulness [16].

The process of creating a product with the user experience in mind is known as user-centred design [17]. Enhancing the User Interface (UI) design and offering the greatest user experience are the goals. It includes a number of phases, including communicating with end users, determining their needs, creating UI prototypes, gathering user input, and working on the feedback. The focus of the entire process is on consumer wants and how the product may effectively satisfy them.

B. The process of developing software

For improved planning and management, the development activities are grouped into discrete phases using the Software Development Process, often known as the Software Development Lifecycle (SDLC) [18]. Requirements, Design, Implementation, Testing, Debugging, Deployment, and Maintenance are the fundamental activities that shape the SDLC paradigm. What distinguishes these actions is how they are carried out, either sequentially or concurrently.

Depending on time and resource constraints, organisations can select from a variety of models while developing software. The development team and management typically make the decision together, taking into account a number of variables such as project-specific requirements, technical, business, and management considerations. Among the most well-known models are the Agile Model [19], Waterfall Model [20], and Iterative Model [21]. To choose the best model for the development, several frameworks are employed, such as the one proposed in [22].

According to Converge research [23], case studies of two industrial companies [24], case studies of multinational corporations [25], and the findings of systematic mapping evaluations from 127 similar studies [26], the combination of Agile and the UCD method has a positive impact on products. The combination of Agile and UCD in software development hasn't been the subject of some of this research. Furthermore, no study has been done to map out the additional Agile difficulties that the UCD method can help with.

III. Methodology

A mixed-method approach was selected in order to obtain a thorough evaluation and benchmark the three distinct applications. The online research, as a quantitative method, helped to operationalise the acceptance-relevant factors, while the qualitative method of thinking aloud monitored the participants' thoughts to understand their action. Figure 1 shows the methodical procedure and the variables under investigation.

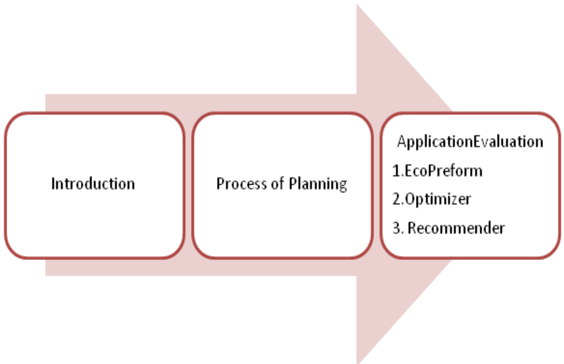

Fig 1. Usability study design for assessing the applications

A. Assessment of AI-Powered Recommender Programs

Three distinct applications for planning and constructing FRP process chains—the "Optimisation App," "Recommender App," and "Eco Preform"—are compared in this study. Since Excel is widely used in industry for task planning, we use Eco Preform as a baseline. On the other hand, the Optimisation App and the Recommender App were created with an emphasis on user-centred design (UCD) and to provide more details on how to employ AI algorithms for planning.

B. Usability

To prevent tiredness or learning effects, we randomly assigned the applications to be examined in the main section. It employed the System Usability Scale (SUS) to evaluate the usability of each application. According to studies acceptance model [27], hedonic motivation (HM), performance expectancy (PE), and usage intention (UI) required to be evaluated in order to provide a more nuanced assessment of the systems' usability. Statements on the intuitivity, comprehensibility, design freedom, and positive sentiment throughout process chain design with each application were included to these things.

Additionally, researchers evaluated automation trust (T) using a scale [28]. Following each task, mental effort was assessed using a 9-point Rating Scale of Mental Effort (RSME). A 7-point Likert scale, ranging from complete agreement to no agreement at all, has to be used to rate every other item.

C. Planning Objectivity and Efficiency

The amount of time required for planning tasks was used to calculate the planning efficiency. Throughout the user study, the study supervisor took time measurements. To ascertain the planning objectivity, participant-designed process chains were recorded and qualitatively contrasted with one another. Technical viability and the uniformity of the levels of automation of the planned process chains were the criterion. Additionally, the process chains' repeatability was investigated in relation to component quality, expenses, and time.

D. Examination

Descriptive data are shown for each of the variables. Since the sample size was tiny, non-parametric tests were employed. Spearman calculates correlations using a significance level of 5% (p) and the correlation coefficient r as an indication. Friedman tests are used to assess mean differences. The effect size was calculated using Cohen's measure. The SUS score ranges from 0 (lowest) to 100 (highest) points and is determined by averaging the ten components. According to meta-studies, SUS scores below 50 are unacceptable, those over 73 are decent, and those over 85 are exceptional.

IV. Empirical Results

The user study's findings are presented in this section, beginning with the general evaluation's descriptive findings for each application. The report of variations in the applications' assessments and the effectiveness comparison come next. The discovery of variables associated with high social acceptance concludes this section.

A. Overall Assessment of Application Types

Various ratings were found when the app types were evaluated. The Recommender App had the highest average rating for every dimension, as seen in Table I, while the Optimisation App came in second. With the exception of two numbers, the Eco Preform's ratings fell below 3.5, which is below the scale's average. The participants only gave favourable evaluations on performance expectancy and trust.

TABLE I. Means And Standard Deviation Of The Variables Under Evaluation

	Recommender		**Optimizer**		**Eco Preform**	
	SD	**Mean**	**SD**	**Mean**	**SD**	**Mean**
Trust	1.01	4.61	.99	4.56	1.52	3.72
Hedonic motivation	.67	5.32	.76	4.97	1.73	2.89
Usage intention	.48	5.28	1.03	4.62	1.59	3.26
Perf. expectancy	.53	5.14	.76	4.98	1.47	3.71

B. Application and System Usability Scale (SUS) comparison

Both the Recommender and the Optimisation App (poor effect with r =.20) and Eco Preform (medium effect with r =.35) had significantly different usage intentions ($\chi^2(2)$ =17.97, p <.001, n=20). The difference in performance expectations between Eco-Preform and the Recommender App was nearly medium (χ^2 (2) =13.52, p=.003, n =20, r=.29). The variables trust (χ^2 (2) =7.52, p=.025, n =20, r =.21) and hedonic motivation (χ^2 (2) =25.16, p <.002, n=20) showed similar patterns between Recommender App and Eco Preform.

The Recommender App received the highest user evaluation with a SUS score of 93.5 points, followed by Eco Preform (50.6) and the Optimisation App (84.4). According to Bangor, Kortum, and Miller, Eco Preform's score is similar to Microsoft Excel's 56.5 points. Summary in summary, every participant thought Eco Preform was less user-friendly than the other two programs, or at least on pace with them.

C. Planning Objectivity & Efficiency

The participants need the least amount of time (02:45 minutes) to independently plan a process chain utilising the Optimisation App in terms of planning efficiency. Participants took an average of 08:52 minutes to plan using Eco Preform, compared to an average of 34 seconds slower (03:23 minutes) while using the Recommender App. A task was viewed as more challenging the longer it took.

The participants' process chains produced using Eco Preform or the Recommender App was all comparable in terms of technological viability and automation levels. The Recommender App also had the highest reproducibility; 15 out of 20 individuals created identical process chain pairs (quality, prices, and duration), whereas Eco Preform and the Optimisation App only produced two identical process chain pairs.

12 of the 20 process chains that were produced using the Optimisation App, however, did not meet the necessary

level of automation and are therefore inconsistent or not technically possible. In these situations, the participants frequently entered the right values, but they failed to click the "optimise" button and did not verify the final result for accuracy (a common "loss of activation error"). The results of the investigation are summarised in Table II.

TABLE II. A COMPARISON OF THE THREE APPS THAT WERE REVIEWED

	Recommender	**Optimizer**	**Eco Preform**
Trust	Largely fulfilled	Largely fulfilled	Partially
User orientation	Fully fulfilled	Largely fulfilled	Not fulfilled
Usage intention	Fully fulfilled	Largely fulfilled	Partially
Planning efficiency	Largely fulfilled	Fully fulfilled	Not fulfilled
Performance expectancy	Fully fulfilled	Fully fulfilled	Partially
Acceptance	Fully fulfilled	Largely fulfilled	Partially
Planning goal	Largely fulfilled	Partially	Partially
Hedonic Evaluation	Fully fulfilled	Largely fulfilled	Partially
Clarity	Largely fulfilled	Largely fulfilled	Not fulfilled
Transparency	Fully fulfilled	Not fulfilled	Partially

D. Estimated Use and Usage Requirements

According to the Technology Acceptance Model, usage intention and subsequent product use are closely related, and the relationship between evaluations and intention to use indicates which features are most important for promoting later use.

For all app types in our study, performance anticipation and intention to use were positively correlated. Eco Preform showed the highest correlation, followed by the Recommender and Optimisation apps. The Optimisation App's purpose to be used was most strongly correlated with hedonic motivation, followed by Eco Preform and the Recommender App. Only the intention to utilise the Optimisation App and Eco Preform could be predicted by trust. There was no discernible association seen in the Recommender App.

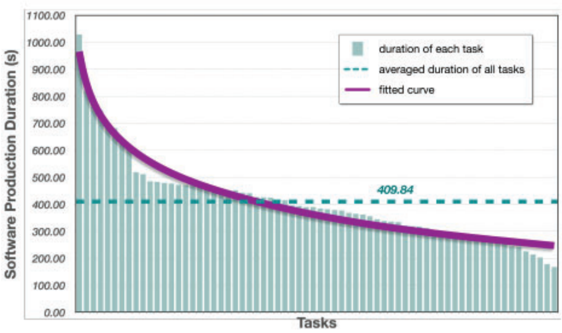

Fig. 2 Distribution of duration

Figure 2 shows the distribution of duration for software development.

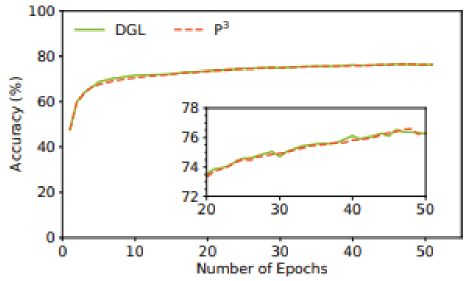

Fig. 3 Accuracy for user centered design

Figure 3 shows the accuracy for user centered design for software development with 79.2%.

V. Conclusion

Three distinct applications with varying capabilities, approaches, and interface designs were given in this research along with an empirical assessment of the applications using a typical user. In conclusion, the analysis demonstrated that AI-based DSSs can help with FRP process chain planning. In order to reward employees and give them the best assistance possible for their work, non-technical factors are just as crucial as technical ones (such as the quality of the outcome). According to consumers, the apps should be simple to understand and utilise, transparent by offering justifications for recommendations, and considerate of how people engage with interactive systems. It discovered that the three apps varied greatly in this respect.

All three applications contain features that participant's value, but the Recommender App receives the highest evaluation. Therefore, it thinks about combining these into one application. The user-friendly Recommender App might incorporate Eco Preform's flexible economic analysis as an expert mode. While professionals have more resources to investigate design alternatives, novices would have an easier time getting started. Additionally, the Recommender App might incorporate the Optimisation App's quick alternative generation: The optimisation model might then produce alternative process chains that are optimal for other target variables (such as costs, quality, and sustainability) when the process chain generation is complete, and encourage planners to assess these as well.

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